

# Aswanth Bindu Ajith

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## 1 PROFESSIONAL SUMMARY

M.Sc. student in High Performance Computing and Quantum Computing with a Mechanical Engineering foundation. Comfortable building high-performance computing, machine learning pipelines, software engineering solutions from low-level numerical code to production web features. Presented optimization findings to a 25-person seminar and received top marks. Reduced simulation runtime by 40

## 2 TECHNICAL SKILLS

HPC Parallel Computing: MPI (basic), OpenMP (basic), GPU-aware computing concepts, Cluster job scheduling (SLURM basics), Performance profiling

Quantum Computing: Qiskit, Cirq (basic), Quantum optimization algorithms, Variational quantum eigen-solver (VQE)

Programming Languages: Python (Advanced, 4+ years), C/C++, JavaScript, Bash/Shell scripting, HTML/CSS

Scientific Computing: NumPy, Pandas, SciPy, Matplotlib, Jupyter, LaTeX

## 3 PROFESSIONAL EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                  |
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| <b>Working Student Software Developer</b>                                                                                                                                                                                                                                                                                                                                                                                                                | Germany          |
| No Border (E-commerce)                                                                                                                                                                                                                                                                                                                                                                                                                                   | Feb 2025 Present |
| <ul style="list-style-type: none"><li>Shipped 5+ frontend features in a 2-week sprint cycle, reducing average ticket age by 30</li><li>Debugged and resolved 20+ production bugs across the React/Vue stack, cutting reported error rates by 25</li><li>Wrote unit tests for 3 core modules, pushing test coverage from 45</li><li>Collaborated with 4-person cross-functional team (design, backend, QA) using GitHub Flow and daily standups</li></ul> |                  |

## 4 EDUCATION

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| <b>M.Sc. High Performance Computing and Quantum Computing</b>                                                                                                           | 2024 Present         |
| Deggendorf Institute of Technology (DIT), Germany                                                                                                                       | Current Average: 2.6 |
| <i>Relevant coursework:</i> Parallel Distributed Computing, Numerical Optimization, Quantum Algorithms, Scientific Software Engineering, High-Dimensional Data Analysis |                      |
| <i>Highlights:</i> Optimization Methods (1.7); System Design Applications (1.7); Software Engineering (2.3); Advanced Mathematics (2.3)                                 |                      |
| <b>B.Tech Mechanical Engineering</b>                                                                                                                                    | 2019 2023            |
| APJ Abdul Kalam Technological University, India                                                                                                                         | CGPA: 6.48           |

## 5 PROJECTS

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| <b>Computing Architecture Optimization Methods</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <ul style="list-style-type: none"><li>Implemented gradient descent and genetic algorithms from scratch to optimize a 50-dimensional engineering cost function</li><li>Benchmarked CPU vs. GPU (CUDA) performance on matrix operations, documenting 8x speedup for large matrices</li><li>Profiled memory bottlenecks using Python's cProfile and reduced peak RAM usage by 35</li><li>Presented findings to a 25-person seminar, receiving top marks on the technical report</li><li>Key metrics: 50-dimensional optimization, 8x GPU speedup, 35</li></ul> |
| <b>Large-Scale Computational Simulation Data Analysis</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <ul style="list-style-type: none"><li>Built a Python simulation engine processing 30,000+ interacting entities using vectorized NumPy operations</li><li>Reduced runtime from 12 minutes to 7 minutes (40</li><li>Identified 3 non-obvious data correlations that shifted the team's modeling assumptions</li></ul>                                                                                                                                                                                                                                         |

- Packaged results into an 18-page LaTeX report with reproducible Jupyter notebooks for peer review
- Key metrics: 30,000+ entities, 40

### **Thermal Analysis of Battery Modules**

- Modeled heat transfer across a 12-cell lithium-ion battery pack using finite difference methods in Python
- Simulated 8 operating scenarios, identifying a thermal runaway risk at  $>45^{\circ}\text{C}$  under sustained 3C discharge
- Proposed a revised cooling channel geometry that improved peak temperature uniformity by 18
- Co-authored a technical report cited by 2 subsequent student projects in the same lab
- Key metrics: 12-cell pack, 8 scenarios, 18

## **6RESEARCH INTERESTS**

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Performance Optimization, High Performance Computing, Scientific Machine Learning, GPU Computing

## **7LANGUAGES**

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English (Professional Working Proficiency), Malayalam (Native), German (A1 (Beginner))